BMP3 Sensor Driver

API Reference — `bmp3_defs.h`

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Overview

This header defines all constants, types, structures, and function-pointer typedefs used by the BMP3 sensor driver (Bosch BMP388 / BMP390). It is the single source of truth for register addresses, bit-field masks, compensation modes, and the main device structure `bmp3_dev`.

Compensation mode is selected at compile time:

- **BMP3_FLOAT_COMPENSATION** (default) — double-precision floating-point output (Pa / °C).
- **BMP3_64BIT_COMPENSATION** — integer-only output (int64_t / uint64_t) for MCUs without FPU.

API Return Codes

Macro	Value	Type	Description
`BMP3_OK`	`0`	**Success**	Operation completed successfully.
`BMP3_E_NULL_PTR`	`-1`	**Error**	A required pointer was NULL.
`BMP3_E_COMM_FAIL`	`-2`	**Error**	Bus (I2C/SPI) communication failed.
`BMP3_E_INVALID_ODR_OSR_SETTINGS`	`-3`	**Error**	ODR/OSR combination is invalid.
`BMP3_E_CMD_EXEC_FAILED`	`-4`	**Error**	Command execution failed.
`BMP3_E_CONFIGURATION_ERR`	`-5`	**Error**	Sensor configuration error detected.
`BMP3_E_INVALID_LEN`	`-6`	**Error**	Invalid data length supplied.
`BMP3_E_DEV_NOT_FOUND`	`-7`	**Error**	No BMP3 found on the bus.
`BMP3_E_FIFO_WATERMARK_NOT_REACHED`	`-8`	**Error**	FIFO watermark level not reached.
`BMP3_W_SENSOR_NOT_ENABLED`	`1`	**Warning**	Requested sensor is not enabled.
`BMP3_W_INVALID_FIFO_REQ_FRAME_CNT`	`2`	**Warning**	Requested FIFO frame count is invalid.
`BMP3_W_MIN_TEMP`	`3`	**Warning**	Temperature below minimum range.
`BMP3_W_MAX_TEMP`	`4`	**Warning**	Temperature above maximum range.
`BMP3_W_MIN_PRES`	`5`	**Warning**	Pressure below minimum range.
`BMP3_W_MAX_PRES`	`6`	**Warning**	Pressure above maximum range.

Sensor Operating Ranges

Parameter	Minimum	Maximum	Unit
Temperature (int)	-4000	8500	×0.01 °C
Temperature (float)	-40.0	85.0	°C

Parameter	Minimum	Maximum	Unit
Pressure (int)	3,000,000	12,500,000	Pa
Pressure (float)	30,000.0	125,000.0	Pa

Enumerations

enum bmp3_intf

Selects the hardware communication interface.

Enumerator	Value	Description
BMP3_SPI_INTF	0	SPI interface selected.
BMP3_I2C_INTF	1	I2C interface selected.

Function Pointer Types

The driver is fully bus-agnostic. The user must supply three callbacks in `bmp3_dev` at initialisation time.

`bmp3_read_fptr_t`

Platform read callback. Reads len bytes starting at reg_addr into read_data.

```
BMP3_INTF_RET_TYPE (*)(uint8_t reg_addr, uint8_t *read_data, uint32_t len, void *intf_ptr)
```

Parameters:

Name	Dir	Description
reg_addr	[in]	8-bit register address to read from.
read_data	[out]	Buffer to receive the data.
len	[in]	Number of bytes to read.
intf_ptr	[in,out]	User-defined interface descriptor pointer.

Returns: 0 (BMP3_INTF_RET_SUCCESS) on success; non-zero on failure.

bmp3_write_fptr_t

Platform write callback. Writes len bytes from read_data starting at reg_addr.

```
BMP3_INTF_RET_TYPE (*)(uint8_t reg_addr, const uint8_t *read_data, uint32_t len, void *intf_ptr)
```

Parameters:

Name	Dir	Description
reg_addr	[in]	8-bit register address to write to.
read_data	[in]	Data buffer to write.
len	[in]	Number of bytes to write.
intf_ptr	[in,out]	User-defined interface descriptor pointer.

Returns: 0 (BMP3_INTF_RET_SUCCESS) on success; non-zero on failure.

bmp3_delay_us_fptr_t

Platform delay callback. Blocks for period microseconds.

```
void (*)(uint32_t period, void *intf_ptr)
```

Parameters:

Name	Dir	Description
period	[in]	Delay duration in microseconds.
intf_ptr	[in,out]	User-defined interface descriptor pointer.

Returns: void

Data Structures

struct bmp3_reg_calib_data

Raw (register-level) trimming coefficients read from sensor NVM. Used internally by the compensation formula.

Type	Member	Description
uint16_t	par_t1	Temperature trim coeff T1
uint16_t	par_t2	Temperature trim coeff T2
int8_t	par_t3	Temperature trim coeff T3
int16_t	par_p1	Pressure trim coeff P1
int16_t	par_p2	Pressure trim coeff P2
int8_t	par_p3	Pressure trim coeff P3
int8_t	par_p4	Pressure trim coeff P4
uint16_t	par_p5	Pressure trim coeff P5
uint16_t	par_p6	Pressure trim coeff P6
int8_t	par_p7	Pressure trim coeff P7
int8_t	par_p8	Pressure trim coeff P8
int16_t	par_p9	Pressure trim coeff P9
int8_t	par_p10	Pressure trim coeff P10
int8_t	par_p11	Pressure trim coeff P11
int64_t	t_lin	Intermediate compensated temperature (linear)

struct bmp3_quantized_calib_data

Double-precision floating-point version of the trim coefficients. Available only when BMP3_FLOAT_COMPENSATION is defined.

Type	Member	Description
double	par_t1/t2/t3	Quantized temperature coefficients
double	par_p1…par_p11	Quantized pressure coefficients
double	t_lin	Intermediate compensated temperature

struct bmp3_calib_data

Top-level calibration container. Members differ by compensation mode.

Type	Member	Description
bmp3_quantized_calib_data	quantized_calib_data	Float coefficients (BMP3_FLOAT_COMPENSATION only)
bmp3_reg_calib_data	reg_calib_data	Raw register coefficients (always present)

struct bmp3_data

Compensated output data. Field types depend on the active compensation mode.

Type	Member	Description
double / int64_t	temperature	Compensated temperature — °C (float) or ×0.01 °C (int)
double / uint64_t	pressure	Compensated pressure — Pa (float) or Pa (int, $\times 10^{■2}$)

struct bmp3_uncomp_data

Raw ADC data straight from the sensor data registers before compensation.

Type	Member	Description
uint64_t	pressure	Uncompensated pressure ADC value
int64_t	temperature	Uncompensated temperature ADC value

struct bmp3_odr_filter_settings

Output data rate and IIR filter configuration.

Type	Member	Description
uint8_t	press_os	Pressure oversampling (BMP3_NO_OVERSAMPLING … BMP3_OVERSAMPLING
uint8_t	temp_os	Temperature oversampling
uint8_t	iir_filter	IIR filter coefficient (BMP3_IIR_FILTER_DISABLE … BMP3_IIR_FILTER_COEFF_
uint8_t	odr	Output data rate (BMP3_ODR_200_HZ … BMP3_ODR_0_001_HZ)

struct bmp3_int_ctrl_settings

Interrupt pin electrical and logical configuration.

Type	Member	Description
uint8_t	output_mode	BMP3_INT_PIN_PUSH_PULL or BMP3_INT_PIN_OPEN_DRAIN
uint8_t	level	BMP3_INT_PIN_ACTIVE_HIGH or BMP3_INT_PIN_ACTIVE_LOW
uint8_t	latch	BMP3_INT_PIN_LATCH or BMP3_INT_PIN_NON_LATCH
uint8_t	drdy_en	BMP3_ENABLE / BMP3_DISABLE data-ready interrupt

struct bmp3_adv_settings

Advanced I2C watchdog timer settings.

Type	Member	Description
uint8_t	i2c_wdt_en	BMP3_ENABLE / BMP3_DISABLE I2C watchdog
uint8_t	i2c_wdt_sel	BMP3_I2C_WDT_SHORT_1_25_MS or BMP3_I2C_WDT_LONG_40_MS

struct bmp3_settings

Master settings structure passed to bmp3_set_sensor_settings().

Type	Member	Description
uint8_t	op_mode	Operating mode: SLEEP, FORCED, or NORMAL
uint8_t	press_en	BMP3_ENABLE / BMP3_DISABLE pressure measurement
uint8_t	temp_en	BMP3_ENABLE / BMP3_DISABLE temperature measurement
bmp3_odr_filter_settings	odr_filter	ODR and IIR filter configuration
bmp3_int_ctrl_settings	int_settings	Interrupt pin configuration
bmp3_adv_settings	adv_settings	I2C watchdog configuration

struct bmp3_sens_status

Sensor readiness flags polled from the STATUS register.

Type	Member	Description
uint8_t	cmd_rdy	Non-zero when device is ready to accept a new command
uint8_t	drdy_press	Non-zero when pressure data is ready
uint8_t	drdy_temp	Non-zero when temperature data is ready

struct bmp3_int_status

Interrupt event flags from the INT_STATUS register.

Type	Member	Description
uint8_t	fifo_wm	FIFO watermark interrupt asserted
uint8_t	fifo_full	FIFO full interrupt asserted
uint8_t	drdy	Data-ready interrupt asserted

struct bmp3_err_status

Error flags from the ERR_REG register.

Type	Member	Description
uint8_t	fatal	Fatal error — sensor requires reset
uint8_t	cmd	Last command failed
uint8_t	conf	Sensor configuration is inconsistent

struct bmp3_status

Aggregate status container returned by bmp3_get_status().

Type	Member	Description
bmp3_int_status	intr	Interrupt event flags
bmp3_sens_status	sensor	Data-ready and command-ready flags
bmp3_err_status	err	Error register flags
uint8_t	pwr_on_rst	Non-zero if a power-on-reset event occurred

struct bmp3_fifo_settings

FIFO enable and configuration written via bmp3_set_fifo_settings().

Type	Member	Description
uint8_t	mode	BMP3_ENABLE / BMP3_DISABLE FIFO
uint8_t	stop_on_full_en	Stop recording when FIFO is full
uint8_t	time_en	Append sensortime frame to FIFO
uint8_t	press_en	Store pressure frames in FIFO
uint8_t	temp_en	Store temperature frames in FIFO
uint8_t	down_sampling	BMP3_FIFO_NO_SUBSAMPLING … BMP3_FIFO_SUBSAMPLING_128X
uint8_t	filter_en	Enable downsampling filter
uint8_t	fwtm_en	Watermark interrupt enable
uint8_t	ffull_en	FIFO-full interrupt enable

struct bmp3_fifo_data

FIFO read-back state. The caller supplies buffer; the driver fills the rest.

Type	Member	Description
uint8_t *	buffer	Caller-supplied data buffer (≥512+4 bytes recommended)
uint16_t	byte_count	Number of bytes returned from the FIFO
uint8_t	req_frames	Number of frames the caller wants to parse
uint16_t	start_idx	Current parse position within buffer
uint8_t	parsed_frames	Number of fully parsed frames
uint8_t	config_err	Non-zero if a configuration-error frame was seen
uint32_t	sensor_time	Sensor timestamp extracted from TIME frame
uint8_t	config_change	Non-zero if a config-change frame was detected
uint8_t	frame_not_available	Non-zero when all available frames have been consumed

struct bmp3_dev

Central device handle. One instance per physical sensor. Must be zero-initialised then populated before calling bmp3_init().

Type	Member	Description
uint8_t	chip_id	Filled by bmp3_init() — 0x50 (BMP388) or 0x60 (BMP390)
void *	intf_ptr	User-supplied pointer passed through to every read/write/delay call
enum bmp3_intf	intf	BMP3_SPI_INTF or BMP3_I2C_INTF — set before bmp3_init()

Type	Member	Description
BMP3_INTF_RET_TYPE	intf_rslt	Last bus error code; checked after every driver call
uint8_t	dummy_byte	SPI dummy byte flag; set automatically by bmp3_init()
bmp3_read_fptr_t	read	User-supplied read callback — must be set before bmp3_init()
bmp3_write_fptr_t	write	User-supplied write callback — must be set before bmp3_init()
bmp3_delay_us_fptr_t	delay_us	User-supplied microsecond delay callback
bmp3_calib_data	calib_data	Calibration coefficients — populated by bmp3_init()

Key Macro Groups

Register Addresses

Base addresses of all BMP3 configuration and data registers.

Macro	Value	Description
BMP3_REG_CHIP_ID	0x00	Chip identification register
BMP3_REG_ERR	0x02	Error status register
BMP3_REG_SENS_STATUS	0x03	Sensor status register
BMP3_REG_DATA	0x04	Burst read start — 6 bytes of P+T data
BMP3_REG_FIFO_DATA	0x14	FIFO data output register
BMP3_REG_PWR_CTRL	0x1B	Power control (mode, press_en, temp_en)
BMP3_REG_OSR	0x1C	Oversampling settings
BMP3_REG_ODR	0x1D	Output data rate selection
BMP3_REG_CONFIG	0x1F	IIR filter selection
BMP3_REG_CALIB_DATA	0x31	Start of NVM calibration data (21 bytes)
BMP3_REG_CMD	0x7E	Command register (reset / FIFO flush)

Power Modes

Written to BMP3_REG_PWR_CTRL bits [5:4].

Macro	Value	Description
BMP3_MODE_SLEEP	0x00	No measurements; lowest power.
BMP3_MODE_FORCED	0x01	Single measurement then returns to sleep.
BMP3_MODE_NORMAL	0x03	Continuous measurement at configured ODR.

Oversampling Settings

Applied to press_os and temp_os fields.

Macro	Value	Description
BMP3_NO_OVERSAMPLING	0x00	×1 — fastest, lowest resolution
BMP3_OVERSAMPLING_2X	0x01	×2
BMP3_OVERSAMPLING_4X	0x02	×4
BMP3_OVERSAMPLING_8X	0x03	×8
BMP3_OVERSAMPLING_16X	0x04	×16
BMP3_OVERSAMPLING_32X	0x05	×32 — slowest, highest resolution

IIR Filter Coefficients

Applied to iir_filter field.

Macro	Value	Description
BMP3_IIR_FILTER_DISABLE	0x00	Filter off
BMP3_IIR_FILTER_COEFF_1	0x01	Coefficient 1
BMP3_IIR_FILTER_COEFF_3	0x02	Coefficient 3
BMP3_IIR_FILTER_COEFF_7	0x03	Coefficient 7
BMP3_IIR_FILTER_COEFF_15	0x04	Coefficient 15
BMP3_IIR_FILTER_COEFF_31	0x05	Coefficient 31
BMP3_IIR_FILTER_COEFF_63	0x06	Coefficient 63
BMP3_IIR_FILTER_COEFF_127	0x07	Coefficient 127 — strongest smoothing

Output Data Rate

Selected ODR values (subset shown).

Macro	Value	Description
BMP3_ODR_200_HZ	0x00	200 Hz
BMP3_ODR_100_HZ	0x01	100 Hz
BMP3_ODR_50_HZ	0x02	50 Hz
BMP3_ODR_25_HZ	0x03	25 Hz
BMP3_ODR_12_5_HZ	0x04	12.5 Hz
BMP3_ODR_1_5_HZ	0x07	1.5 Hz
BMP3_ODR_0_1_HZ	0x0B	0.1 Hz
BMP3_ODR_0_001_HZ	0x11	0.001 Hz — slowest

FIFO Sub-sampling

Down-sampling factor stored in bmp3_fifo_settings.down_sampling.

Macro	Value	Description
BMP3_FIFO_NO_SUBSAMPLING	0x00	Every sample stored
BMP3_FIFO_SUBSAMPLING_2X	0x01	Every 2nd sample
BMP3_FIFO_SUBSAMPLING_4X	0x02	Every 4th sample
BMP3_FIFO_SUBSAMPLING_8X	0x03	Every 8th sample
BMP3_FIFO_SUBSAMPLING_128X	0x07	Every 128th sample

FIFO Header Bytes

Frame-type identifiers at the start of each FIFO record.

Macro	Value	Description
BMP3_FIFO_TEMP_PRESS_FRAME	0x94	Frame contains both temperature and pressure
BMP3_FIFO_TEMP_FRAME	0x90	Temperature-only frame
BMP3_FIFO_PRESS_FRAME	0x84	Pressure-only frame
BMP3_FIFO_TIME_FRAME	0xA0	Sensor-time frame
BMP3_FIFO_ERROR_FRAME	0x44	Error frame
BMP3_FIFO_CONFIG_CHANGE	0x48	Configuration-change frame
BMP3_FIFO_EMPTY_FRAME	0x80	Empty / no more data

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